

4(a)(i)	<table border="1" data-bbox="342 248 1247 576"> <thead> <tr> <th>statement</th><th>meiosis</th><th>mitosis</th></tr> </thead> <tbody> <tr> <td>chromosome number is maintained</td><td>×</td><td>✓</td></tr> <tr> <td>homologous chromosomes pair up</td><td>✓</td><td>×</td></tr> <tr> <td>sister chromatids separate</td><td>✓</td><td>✓</td></tr> <tr> <td>occurs in prokaryotes</td><td>×</td><td>×</td></tr> </tbody> </table> all 4 rows correct = 2 marks 2/3 rows correct = 1 mark A yes / no for ✓ / ×	statement	meiosis	mitosis	chromosome number is maintained	×	✓	homologous chromosomes pair up	✓	×	sister chromatids separate	✓	✓	occurs in prokaryotes	×	×	2
statement	meiosis	mitosis															
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4(a)(ii)	any four from: <i>similarities</i> 1 (both / meiosis) produce 4 (cells) ; 2 (both / meiosis) halve chromosome number / haploid / n / one set of chromosomes ; 3 (both / meiosis) produce genetically different cells ; <i>Differences – must be comparative statements</i> 4 sperm is a gamete whereas a pollen grain is, not a gamete / a gametophyte ; 5 sperm cell has one (haploid) nucleus whereas pollen grains contain two (haploid) nuclei ; 6 sperm mitosis then meiosis whereas pollen grain cells meiosis then mitosis ; 7 <i>ref. to</i> primary spermatocyte versus (pollen grain) mother cell ;	4															

Question	Answer	Marks
4(b)(i)	RR dd PP pale pink petals, red eye ; Rr Dd Pp dark pink petals, red eye ; rr Dd Pp white petals, no red eye / white centre ; RR dd pp white petals, red eye ;	4
4(b)(ii)	<i>any four from:</i> 1 base, substitution / insertion / deletion ; 2 frameshift / described ; 3 changes, primary structure / amino acid sequence / primary sequence ; 4 changes, tertiary structure / 3D shape ; 5 STOP codon ; 6 shortened, protein / polypeptide ; 7 (protein / enzyme) unable to bind to, substrate / receptor / DNA ; 8 (so biosynthetic) pathway does not function / AW ;	4
4(b)(iii)	<i>any one from:</i> 1 is a binding site ; 2 is a, regulatory region / promoter / enhancers / stop codon / telomere ;	1